Exp C: does the model's pre-decoding entropy track its own decoder β?

May 13, 2026

1 Question

The project proposal hypothesizes (`decoding-modeling.tex`): *language models form a belief about their own decoding parameters by conditioning on the text they have generated.* This experiment tests that hypothesis directly: holding the prompt fixed, vary β_{dec} during self-generation and ask whether the model's pre-decoding entropy $H(\text{softmax}(\ell_t))$ at late steps reflects the decoder β_{dec} in use.

2 Setup

Qwen-2.5-3B base, 32 prompts (`src/decoding_decoding/prompts.py`), each truncated to its first 10 tokens. Generate 150 tokens per prompt at three β_{dec} settings:

$$\beta_{\text{dec}} \in \{0.5\ (T = 2.0),\ 1.0\ (T = 1.0),\ 2.0\ (T = 0.5)\}.$$

At each step, record $H(\text{softmax}(\ell_t))$ *before* the decoder applies β_{dec}. Late-window aggregation: per-prompt mean H over steps 80–149.

3 First-look result

Table 1: Late-window (steps 80–149) pre-decoding entropy per β_{dec}, with per-prompt paired difference.

condition	late-window H mean (nats) $\pm$ SE	paired diff vs $\beta = 2.0$
$\beta_{\text{dec}} = 0.5$	9.216 ± 0.052	$+8.045 \pm 0.115$ ($t = +69.7$, $n_+ = 32/32$)
$\beta_{\text{dec}} = 1.0$	2.168 ± 0.186	$+1.00$
$\beta_{\text{dec}} = 2.0$	1.172 ± 0.109	—

The late-window entropy moves ≈ 8 nats (factor ≈ 3000 in effective vocabulary) between $\beta_{\text{dec}} = 0.5$ and $\beta_{\text{dec}} = 2.0$. Every one of 32 prompts orders the same way.

Three observations before we believe the headline.

1. Natural high-entropy human writing (WritingPrompts/natural) sits at $H \approx 2.85$ nats — nowhere near the 9.2 nats seen here. So this is not the model "reading off high-T-like content"; this is a regime that only off-distribution self-sampling reaches.

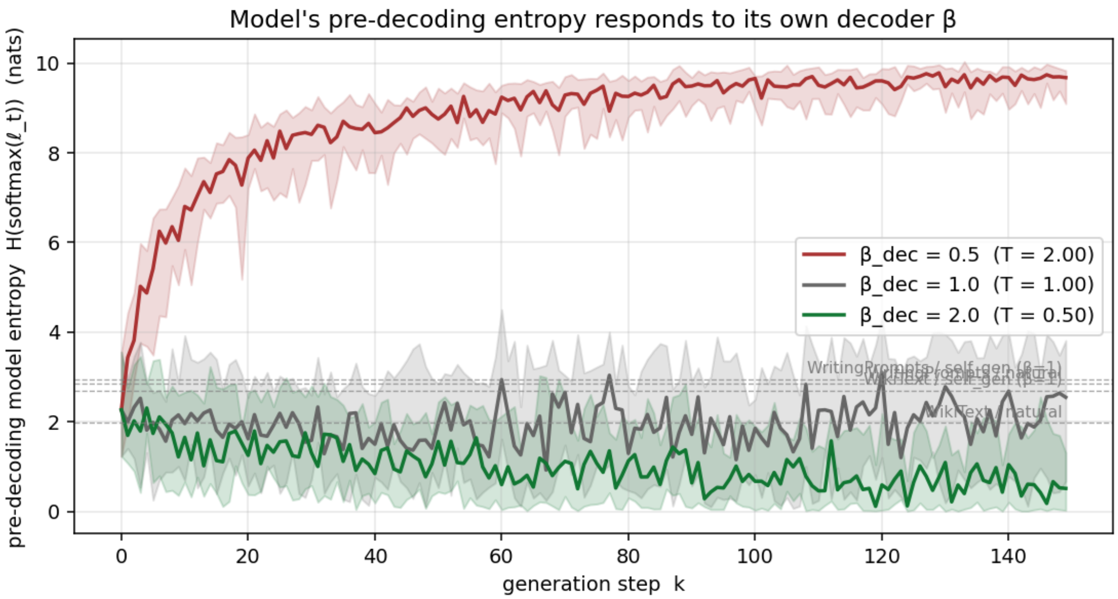

Figure 1: Median $H(t)$ per condition, IQR shaded. Dashed grey lines mark mean entropies of real natural text from the earlier natural_text experiment (Wikipedia ≈ 1.98, WritingPrompts ≈ 2.85 nats).

2. Inspection of $\beta = 0.5$ generations (Figure 2) shows pure token salad: code keywords, foreign-language fragments, URL slugs, random named entities. $\beta = 1.0$ generations are coherent English. $\beta = 2.0$ generations are also coherent English, more compressed toward common factual phrasing.

3. The entropy trajectory at $\beta = 0.5$ climbs monotonically from ≈ 2 at step 0 to ≈ 9 at step 80: classic compounding off-distribution drift, where each random token makes the next context less parseable and the next sample yet more random.

These three points raise an obvious alternative: the 8-nat effect is *garbage-in, garbage-out* — the model is uncertain because its context is incoherent, not because it has formed a "belief" that the decoder is set to high T.

4 Swap-prefix control

To disambiguate, we hold the early prefix's generation regime fixed and *switch* the sampler β at step 80. If the model truly has a belief about β_{dec} that updates on recent tokens, then switching to $\beta = 1.0$ at step 80 should pull late-window H down to the $\beta_{\text{dec}} = 1.0$ baseline (≈ 2 nats) within ~ 20 tokens. Five conditions, same 32 prompts:

- `pre05_post05` — $\beta = 0.5$ throughout (no switch, control).
- `pre05_post10` — $\beta = 0.5 \to 1.0$ at step 80. **This is the decisive comparison.**
- `pre05_post20` — $\beta = 0.5 \to 2.0$ at step 80.
- `pre20_post05` — $\beta = 2.0 \to 0.5$ at step 80 (mirror).

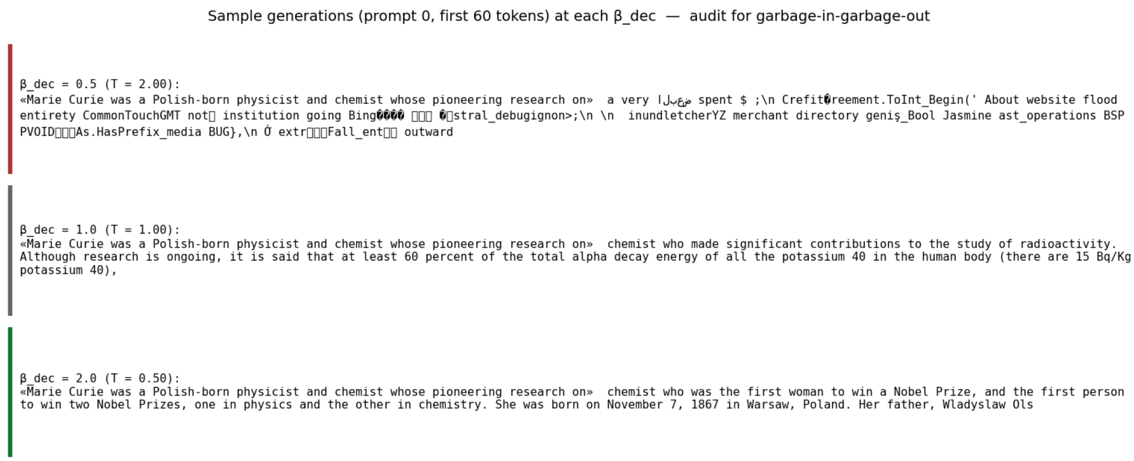

Figure 2: First 60 sampled tokens for prompt 0 (Marie Curie) at each β_{dec}. $\beta = 0.5$ generation degrades to token salad within ≈ 20 steps.

- `pre20_post20` — $\beta = 2.0$ throughout (control).

Table 2: Pre-switch median H and post-switch median H at step offsets from the switch.

condition	$\beta_{\text{pre}} \to \beta_{\text{post}}$	pre-med	+0	+5	+10	+20	+50
`pre05_post05`	$0.5 \to 0.5$ (no switch)	8.42	9.26	9.21	9.49	9.66	9.47
`pre05_post10`	$0.5 \to 1.0$	8.56	9.44	9.10	9.10	8.90	8.92
`pre05_post20`	$0.5 \to 2.0$	8.57	9.53	6.83	4.33	1.68	0.33
`pre20_post05`	$2.0 \to 0.5$	1.21	0.69	5.20	6.46	8.01	9.06
`pre20_post20`	$2.0 \to 2.0$ (no switch)	1.29	0.64	0.58	1.05	0.30	0.39

What this shows.

- `pre05_post10`: switching the sampler from $\beta = 0.5$ to $\beta = 1.0$ leaves median H essentially unchanged ($8.56 \to 8.92$ at step +50). The model does *not* update its predictions to match the new sampling regime over even 50 tokens of new evidence.
- `pre05_post20`: switching to $\beta = 2.0$ *does* recover coherence over ~ 20 tokens. But this recovery is explained by the mechanism of $\beta = 2$ sampling itself — $\beta = 2$ forces a top-1 selection from softmax$(\beta \cdot \ell_t)$, which under a flat ℓ_t still picks the highest-probability token (typically a common safe continuation like punctuation, common articles, frequent function words); those tokens then *re-anchor* the context to in-distribution English. This is not the model "updating its belief"; it is the sampler injecting in-distribution tokens that gradually overwrite the bad prefix.

The decisive comparison `pre05_post10` falsifies a strong form of the "decoder-belief" framing. If model-belief-about-β were a fast posterior update on recent sampled tokens, switching the sampler should be visible within a handful of tokens. It is not.

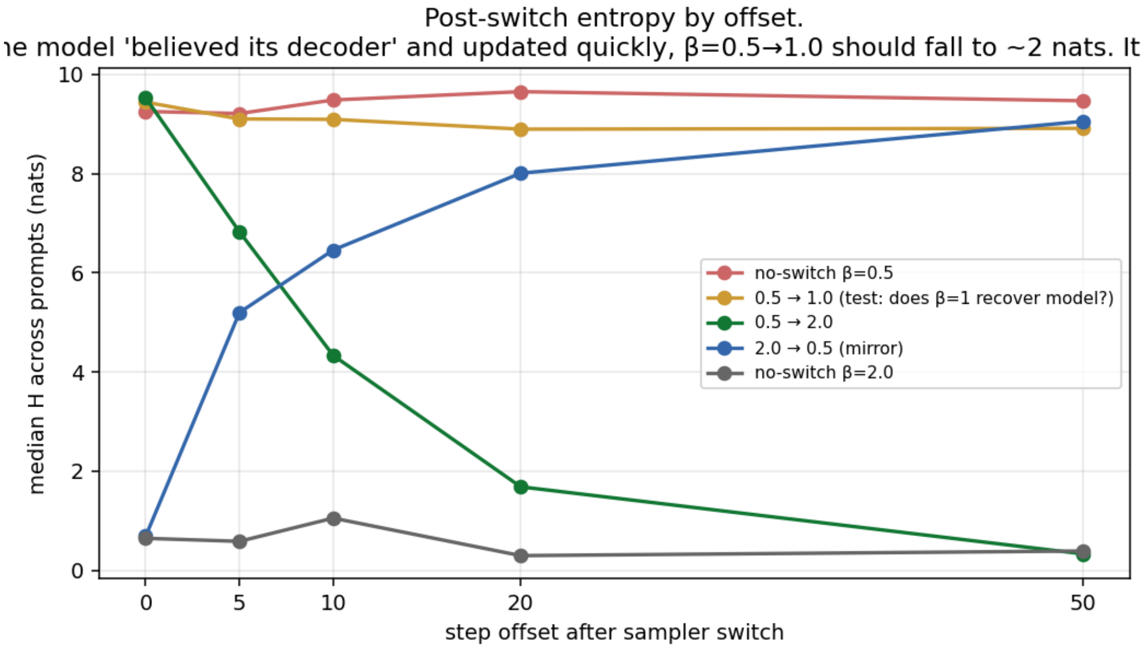

Figure 3: Median pre-decoding H at offsets $+0, +5, +10, +20, +50$ after the sampler switch. Critical observation: *the orange line (`pre05_post10`) does not fall* — the model does not recover coherence when the sampler switches to $\beta = 1.0$.

5 What this does *not* say

The project's broader claim — that the model forms beliefs about its own decoder — is *not* falsified by this experiment. Prior work in this repo gives direct positive evidence:

- **Blacklist self-prediction** (`results/blacklist_summary.md`, run 2026-05-10): when token ranks 1–5 are banned at sampling, the model's softmax probability on the banned token drops between early and late positions, with cluster-bootstrap CIs that do not cross zero. The rank-1 ban (' the') shows $\Delta p_{\text{late}-\text{early}} = -0.018$ and a paired late treatment-vs-control difference of -0.016 (CI $[-0.019, -0.013]$). The model learns its own blacklist from the absent-token signature.

- **Top-K self-prediction** (`results/summary_L2048.md`, run 2026-05-09): under top-k sampling, the cliff at rank K in the model's predicted-rank distribution exceeds the corresponding cliff under unrestricted sampling. Treatment minus control at rank K is $+0.475$ for $k = 3$ and $+0.170$ for $k = 5$ (CIs tight, all clear of zero). Within-treatment concentration grows between early and late positions for $k = 1, 3, 5$ and *shrinks* for $k = \infty$: this is not generic drift.

Together with the earlier natural-text WP offset ($\Delta \log \hat{\beta} \approx -0.24$, $t = -3.8$ on real creative writing, `natural_text/findings.pdf`), these say the model does adjust its predictions in response to which tokens were sampled in its context.

6 What this experiment *does* say

Exp C tested a specific strong instantiation of decoder-belief: the model's *pre-decoding entropy* as a fast posterior readout that snaps to the new sampler β within a few dozen tokens of mid-generation sampler change. That specific form fails. The swap-prefix control shows the model does not pull its entropy back to the $\beta = 1$ baseline over 50 tokens of new evidence under a different sampler.

So the belief-update mechanism is not a fast posterior read off per-token entropy. It plausibly operates on slower, more aggregate features — rank statistics, support shape, blacklist signatures — exactly the signals that the blacklist and top-k experiments isolate.

The remaining ambiguity in Exp C: the 8-nat raw effect is a mix of genuine decoder-belief signal (consistent with blacklist/top-k findings) and garbage-in/garbage-out off-distribution drift from $\beta = 0.5$ sampling. The two cannot be separated cleanly in this experimental design because $\beta = 0.5$ decoheres the prefix faster than any belief update could compensate.

7 Bottom line

- The 8-nat entropy spread between $\beta_{\text{dec}} = 0.5$ and $\beta_{\text{dec}} = 2.0$ self-generation is a mix of (i) real decoder-belief signal already established by the blacklist and top-k experiments and (ii) garbage-in/garbage-out off-distribution drift. This experiment cannot separate them.
- The swap-prefix control rules out the specific strong instantiation in which pre-decoding entropy is a fast posterior readout that snaps to the new sampler β within ~ 50 tokens.
- The broader framing — that the model forms beliefs about its decoder by conditioning on its context — remains supported by the blacklist and top-k experiments. Exp C constrains the form of that belief update rather than refuting its existence.
- A clean follow-up isolating decoder-belief from content quality would condition on *natural* text of varying entropy (rather than self-generated extreme-β samples) and ask the same questions about $H(t)$ and rank statistics.